Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 07-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase 4 Study to Evaluate Safety and Effectiveness of Luspatercept (ACE-536) for the Treatment of Anemia Due to IPSS-R Very Low, Low, or Intermediate Risk Myelodysplastic Syndromes (MDS) With Ring Sideroblasts Who Require Red Blood Cell Transfusions in Participants Who Have Had Unsatisfactory Response to or Are Ineligible to Erythropoietin Based Therapy and in Participants With Transfusion Dependent Anemia Due to Beta-Thalassemia **Short Title/Acronym:** Luspatercept Safety and Effectiveness in TD Anemia (MDS & β-Thal) in India **Protocol Number:** CA056-023 **NCT Number:** NCT05891249 **Posting ID:** 716558171765213578 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Luspatercept (ACE-536 / Reblozyl / BMS-986346) **Phase of Development:** Phase 4 **Trial Status:** Active, Not Recruiting **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and effectiveness of luspatercept in participants who require regular red blood cell (RBC) transfusions due to β-thalassemia and myelodysplastic syndromes (MDS) in India. **Secondary Objectives:** To evaluate red blood cell transfusion independence (RBC-TI), reduction in RBC transfusion burden, and long-term overall safety and survival outcomes.

2.2 Background & Rationale To address the significant medical burden of transfusion-dependent anemia in patients with β-thalassemia and Myelodysplastic Syndromes (MDS) in India. Regular blood transfusions lead to iron overload and numerous complications. Luspatercept acts as an erythroid maturation agent to promote late-stage red blood cell production. This post-marketing Phase 4 study provides real-world effectiveness and safety data specifically evaluating the clinical benefit and transfusion burden reduction within the Indian healthcare setting.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Active, Not Recruiting

3.2 Planned Enrollment & Duration Target N: Approximately 85 participants **Number of Sites:** Multicenter study in India (including sites in Bangalore, Chandigarh, Delhi, and Kolkata) **Estimated Study Start:** 05-Jun-2023 **Estimated Primary Completion:** Q4 2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

β-Thalassemia Cohort: 1. Documented diagnosis of β-thalassemia or hemoglobin (Hb E/β-thalassemia). (β-thalassemia with mutation and/or multiplication of alpha [α] globin is allowed). 2. Regularly transfused, defined as 6 RBC units to 20 RBC units in the 24 weeks prior to enrollment and no transfusion-free period for > 35 days during that period.

MDS-RS Cohort: 1. Participant has documented diagnosis of MDS according to World Health Organization (WHO) (2016)/French-American-British FAB classification that meets revised International Prognostic Scoring System (IPSS-R) classification of very low, low, or intermediate risk disease and the following criteria: i) Ring Sideroblasts (RS) $\ge 15\%$ of erythroid precursors in bone marrow. If the SF3B1 mutation is present, RS $\ge 5\%$ will be included. ii) Less than 5% blasts in bone marrow and < 1% peripheral blood blasts. iii) Peripheral blood white blood cell (WBC) count < 13,000/ microliters (μL). 2. If the participant was previously treated with erythropoiesis-stimulating agents (ESAs) or granulocyte colony-stimulating factor (G-CSF)/granulocyte-macrophage colony-stimulating factor (GM-CSF), both agents must have been discontinued $\ge$ 4 weeks prior to the date of enrollment.

4.2 Exclusion Criteria

β-Thalassemia Cohort: 1. A diagnosis of Hb S/β-thalassemia or α-thalassemia (for example, Hemoglobin H). 2. Deep vein thrombosis (DVT) or stroke requiring medical intervention ≤ 24 weeks prior to enrollment. 3. Use of chronic anticoagulant therapy is excluded unless the treatment stopped at least 28 days prior to enrollment. Anticoagulant therapies used for prophylaxis for surgery or high-risk procedures as well as low-molecular-weight (LMW) heparin for superficial venous thrombosis and chronic aspirin are allowed. 4. Cytotoxic agents or immunosuppressants or immunomodulatory drugs (IMiDs) ≤ 28 days prior to enrollment (i.e., antithymocyte globulin or cyclosporine or thalidomide).

MDS-RS Cohort: 1. MDS associated with del 5q cytogenetic abnormality. 2. Secondary MDS, that is, MDS that is known to have arisen as the result of chemical injury or treatment with chemotherapy and/or radiation for other diseases. 3. Participant has known clinically significant anemia due to iron, vitamin B12, or folate deficiencies; autoimmune or hereditary hemolytic anemia; or gastrointestinal bleeding. 4. Iron deficiency to be determined by serum ferritin ≤ 15 micrograms per liter (μg/L) and additional testing if clinically indicated (for example, calculated transferrin saturation [iron/total iron binding capacity $\le 20\%$] or bone marrow aspirate [BMA] stain for iron).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Luspatercept administered at a specified starting dose (typically 1.0 mg/kg), with potential titration based on response and tolerability, given once every 3 weeks (21-day cycles). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or loss of clinical benefit.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for anemia, including iron chelation therapy, prophylactic LMW heparin for superficial venous thrombosis/ surgery, and chronic aspirin. **Prohibited:** Cytotoxic agents, systemic immunosuppressants (e.g., cyclosporine, antithymocyte globulin), immunomodulatory drugs (e.g., thalidomide), and erythropoiesis-stimulating agents (ESAs) or G-CSF/GM-CSF within $\ge$ 4 weeks of enrollment. Chronic anticoagulant therapies without allowable indications must be stopped 28 days prior.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical/Transfusion History, Labs, IPSS-R Scoring
Baseline	Day 1	First Subcutaneous Dose, Safety Labs, Vitals
Treatment Cycles	Every 21 Days ± 3	SC Injection of Luspatercept, Efficacy (RBC burden) Assessment, AE Monitoring
End of Treatment	Post-discontinuation	Final Safety Check, IP Accountability, Exit Clinical Assessment
Follow-Up	Every 12 Weeks	Long-term Safety, RBC Transfusion Status Tracking

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **β-Thal Cohort:** Number of participants with treatment-related adverse events (AEs) of grade 3 or higher. 2. **MDS-Ring Sideroblasts (RS) Cohort:** Number of participants with treatment-related AEs of grade 3 or higher.

7.2 Secondary Endpoints 1. **β-Thal Cohort:** Percentage of participants who achieved red blood cell (RBC) transfusion burden reduction ($\ge 33\%$ reduction from baseline) with a reduction of at least 2 red cell units compared to the 12-week interval prior to enrollment. 2. **β-Thal Cohort:** Percentage of participants who achieved RBC transfusion burden reduction of at least 33% from baseline during any 12-week interval with a reduction of at least 2 red cell units compared to the 12-week interval prior to enrollment. 3. **MDS-RS Cohort:** Percentage of participants who achieved Red Blood Cell Transfusion Independence (RBC-TI) during any consecutive 56-day period.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from informed consent through the 12-week safety follow-up period post-last dose, focusing on treatment-related grade 3 or higher AEs, thromboembolic events, and hypertension. **Serious Adverse Events (SAEs):** Must be reported to the sponsor (BMS) and relevant Indian regulatory authorities within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Regular safety reviews conducted by an internal BMS safety governance team.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for baseline demographics and efficacy evaluations; Safety Set for all participants receiving at least one dose of luspatercept. **Sample Size Justification:** A sample of approximately 85 participants provides adequate descriptive power to evaluate the real-world safety and effectiveness within the Indian sub-population. **Handling of Missing Data:** Participants withdrawing prior to achieving an efficacy milestone are considered non-responders for categorical transfusion burden endpoints.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and Indian regulatory requirements (CDSCO). **Monitoring Plan:** Standard risk-based onsite and remote monitoring at all participating medical centers in India. **Audit Trail:** Utilizing secure EDC (Electronic Data Capture) systems with continuous audit trails and query generation processes.

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11. Study Identifiers & Governance

Primary ID: CA056-023 **Registry Number:** NCT05891249 **Posting ID:** 716558171765213578 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate the Safety and Effectiveness of Luspatercept for the Treatment of Transfusion-dependent (TD) Anemia Associated With Myelodysplastic Syndromes (MDS) & Beta-thalassemia (β-Thal) in India **Official Regulatory Title:** A Phase 4 Study to Evaluate Safety and Effectiveness of Luspatercept (ACE-536) for the Treatment of Anemia Due to IPSS-R Very Low, Low, or Intermediate Risk Myelodysplastic Syndromes (MDS) With Ring Sideroblasts Who Require Red Blood Cell Transfusions in Participants Who Have Had Unsatisfactory Response to or Are Ineligible to Erythropoietin Based Therapy and in Participants With Transfusion Dependent Anemia Due to Beta-Thalassemia

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12. Clinical Framework

Disease Areas: Anemia **Preferred UMLS Name:** Transfusion-Dependent Anemia **Primary Condition:** Preleukemia / Myelodysplastic Syndromes (MDS) and β-Thalassemia **Diagnosis Threshold:** Clinically symptomatic anemia requiring 6 to 20 RBC units transfused over the prior 24 weeks. **Medical Methodology:** Erythroid Maturation Agent (EMA) Therapy **Optimization Target:** Significant reduction in RBC transfusion burden and achievement of transfusion independence.

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13. Study Procedures & Methodology

Management Pathway: Ongoing subcutaneous administration of luspatercept aligned with standard institutional management for transfusion-dependent anemia. **Education Protocol (HCP):** Sponsor-provided training on subcutaneous administration, drug handling, and AE monitoring specific to Indian healthcare sites. **Education Protocol (Patient):** Counseling on deep vein thrombosis (DVT) risks, strict AE reporting, and accurate transfusion log-keeping. **Quality Audit Mechanism:** Ongoing remote source data verification focusing on hemoglobin lab trends and confirmed RBC unit dispenses.

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14. Observation & Follow-Up Schedule

Total Duration: Dependent on individual benefit, with long-term safety tracking **Onsite Visit Cadence:** Every 3 weeks for treatment dosing and evaluations **Remote Monitoring Frequency:** Intermittent per site standard practice. **Checklist Review Interval:** Routine check of hematology labs and transfusion logs prior to every 21-day administration cycle.

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | |**

Clinical Operations Lead | [Name] | ___ | ***[DD-MMM-YYYY]* | |** ***Lead Statistician* | *[Name]* |** _____ | [DD-MMM-YYYY] |